

Project Report: Automotive Cybersecurity Service Prediction System

1. Data Understanding Report

1.1 Problem Statement

The objective of this project is to build a machine learning model that predicts which cybersecurity subscription package (Cybersecurity_Service) a vehicle/customer is most likely to adopt.

Business Context

Modern vehicles are becoming Software Defined Vehicles (SDVs) and Connected Cars, with features such as cloud connectivity, mobile app integration, ADAS, OTA updates, and remote diagnostics. As vehicles become more connected, cybersecurity requirements increase. A predictive system can help automotive OEMs:

- Identify the right cybersecurity package for a vehicle profile early in the sales cycle
- Understand the key drivers of cybersecurity package adoption
- Design targeted product and marketing strategies for each package tier

1.2 Objective

- Predict the cybersecurity package adopted (5-class classification)
- Identify key vehicle factors contributing to package adoption
- Provide actionable recommendations to product/sales teams

1.3 Target Variable

- Cybersecurity_Service
- 5 classes: Basic Monitoring, Threat Detection, OTA Security, Vulnerability Management, Premium Cyber Suite

1.4 Input Variables

The dataset contains 15 columns (14 input features + 1 target). Key input variables include:

Numerical Features:

- Year
- Price
- Mileage
- Engine_CC
- Seating_Capacity
- ADAS_Level
- Vehicle_Age

Categorical Features:

- Brand
- Model
- Fuel_Type
- Transmission

- Vehicle_Segment
- Connected_Car
- OTA_Update_Enabled

1.5 Initial Data Checks

The following checks were performed on load:

- df.shape → Dataset dimensions: 3,000 rows × 15 columns
- df.info() → Data types and null value summary
- df.describe() → Statistical summary (mean, std, min, max, quartiles)
- df.head() → Sample row inspection

Observations

- No null / missing values detected in the dataset
- Mix of numerical and categorical (object) variables
- Target variable has moderate class imbalance: Premium Cyber Suite is the smallest class (9.8%)

1.6 Data Cleaning Summary

No constant/ID columns were found requiring removal; all 14 input columns carry potential predictive value.

Null Value Handling

Column Type	Strategy
Categorical (text)	Mode Imputation
Numerical	Median Imputation

Note: No null values were present in this dataset; imputation logic was implemented as a defensive best-practice pattern.

Duplicate Handling

- Checked using df.duplicated().sum()
- Removed using df.drop_duplicates()
- Result: 0 duplicate rows found

Error Value Handling

- Scanned all columns for values like '?', '@', '#', 'NA', 'N/A', 'none'
- Replaced any found error values with NaN, then re-applied imputation
- Result: no error values detected in this dataset

Note on Outliers: IQR-based outlier removal was intentionally NOT applied. In automotive data, a very high Price or Engine_CC may represent a valid luxury vehicle rather than a data error, so all data points were retained for model training.

1.7 Exploratory Analysis Summary

Target Distribution

Package	Count	Share
Threat Detection	716	23.9%
OTA Security	704	23.5%
Vulnerability Management	675	22.5%
Basic Monitoring	612	20.4%
Premium Cyber Suite	293	9.8%

Plot saved: output_plots/01_service_distribution.png

Univariate Analysis

- Histograms plotted for: Price, Mileage, Engine_CC, Year, Vehicle_Age, ADAS_Level
- Bar charts plotted for: Brand, Fuel_Type, Transmission, Vehicle_Segment, Connected_Car, OTA_Update_Enabled
- Plots saved: output_plots/02_numerical_univariate_analysis.png | 03_categorical_univariate_analysis.png

Bivariate Analysis

- Cybersecurity package mix analysed by Connected_Car, OTA_Update_Enabled, Vehicle_Segment, ADAS_Level
- Numerical feature distributions split by Cybersecurity_Service (overlaid density plots)
- Plots saved: output_plots/04_service_by_key_features.png | 05_numerical_features_by_service.png

Key Observations

- Connected vehicles adopt far more advanced packages than non-connected vehicles
- OTA-enabled vehicles show much higher Premium Cyber Suite adoption (21.9% vs 0.3% for non-OTA)
- Higher ADAS_Level strongly correlates with premium package adoption (32.7% at ADAS Level 3)
- Luxury and Premium vehicle segments show the highest advanced-package adoption rates

2. Feature Engineering Documentation

2.1 Target Encoding

Column	Original Values	Encoded Values	Method
Cybersecurity_Service	5 package names	0-4	LabelEncoder

2.2 Categorical Encoding

All nominal categorical columns were transformed using One-Hot Encoding:

Type	Columns	Method
Nominal	Brand, Model, Fuel_Type, Transmission, Vehicle_Segment, Connected_Car, Onboard_Diagnostic, App_Updated, Email, drop_first=	One-Hot Encoding

Note: drop_first=True in One-Hot Encoding eliminates the dummy variable trap and reduces multicollinearity.

2.3 Feature Transformation

- StandardScaler applied to all input features after train-test split
- Scaler fitted only on training data (X_train) to prevent data leakage into the test set
- Both X_train_scaled and X_test_scaled produced for model input

2.4 Final Feature Set

- Input matrix X: shape (3000, 48) after one-hot encoding expansion
- Target vector y: Cybersecurity_Service (5 encoded classes)
- Train set: 80% of data (2,400 rows) | Test set: 20% of data (600 rows)
- stratify=y_encoded ensures both splits maintain the same class distribution

3. Model Development and Evaluation Report

3.1 Dataset Splitting

Split	Size	Variables
Training set	80% (2,400 rows)	X_train, y_train
Test set	20% (600 rows)	X_test, y_test

- train_test_split used with random_state=42 for reproducibility
- stratify=y_encoded applied to preserve class distribution in both splits

3.2 Models Implemented

- **1. Logistic Regression** - baseline linear multi-class classifier, max_iter=2000
- **2. Decision Tree** - handles non-linear relationships, highly explainable, max_depth=10
- **3. Random Forest** - ensemble of 200 trees, reduces overfitting, max_depth=15; primary source of feature importance

3.3 Model Training

- All models trained using model.fit(X_train_scaled, y_train)
- StandardScaler applied before training to normalize all features to the same scale

3.4 Cross Validation

- 5-Fold Stratified Cross-Validation applied to all models on the training set
- StratifiedKFold(n_splits=5, shuffle=True, random_state=42)
- Metrics evaluated per fold (weighted): Accuracy, Precision, Recall, F1-Score

Model	Accuracy	Precision	Recall	F1 Score
Decision Tree	0.9700	0.9703	0.9700	0.9700
Logistic Regression	0.9571	0.9573	0.9571	0.9570
Random Forest	0.9129	0.9158	0.9129	0.9131

3.5 Hyperparameter Tuning

GridSearchCV (cv=3, scoring='f1_weighted') was applied to find optimal hyperparameters:

Model	Parameter Tuned	Best Value
Logistic Regression	C	10
Decision Tree	max_depth, min_samples_split	10, 2
Random Forest	n_estimators, max_depth	100, 15

3.6 Model Evaluation Metrics

Metric	Description
Accuracy	Overall percentage of correct predictions
Precision (weighted)	Of all predicted packages, how many were correct, weighted by class support
Recall (weighted)	Of all actual packages, how many were correctly identified

F1-Score (weighted)	Harmonic mean of Precision and Recall (primary metric)
Confusion Matrix	Class-by-class breakdown of correct/incorrect predictions

Weighted F1-Score is the primary selection metric because the dataset has moderate class imbalance across 5 cybersecurity packages.

3.7 Results Summary (Test Set)

Model	Accuracy	Precision	Recall	F1-Score
Decision Tree	97.8%	97.8%	97.8%	97.8%
Logistic Regression	96.8%	96.9%	96.8%	96.8%
Random Forest	94.7%	94.7%	94.7%	94.7%

3.8 Model Selection

Selected Model: Decision Tree (highest weighted F1-Score = 0.978 on the test set)

- Highest F1-Score on the held-out test set
- Consistent performance across cross-validation and final test evaluation
- Highly interpretable - decision paths can be shown to non-technical stakeholders

4. Feature Importance Analysis

4.1 Method

Feature importances extracted from the trained Random Forest model using `rf.feature_importances_` (mean decrease in impurity across all trees). Top 15 features visualised and saved to `output_plots/07_feature_importance.png`.

4.2 Top Features Driving Cybersecurity Package Adoption

Rank	Feature	Importance	Interpretation
1	ADAS_Level	0.1715	Higher ADAS sophistication drives premium adoption
2	Price	0.1533	Higher-priced vehicles adopt more advanced packages
3	Connected_Car (Yes)	0.1368	Connectivity is a major adoption trigger
4	OTA_Update_Enabled (Yes)	0.1340	OTA capability signals need for stronger security
5	Mileage	0.0581	Usage pattern has moderate influence
6	Vehicle_Segment_Luxury	0.0332	Luxury segment favors premium packages
7	Engine_CC	0.0314	Larger engines correlate with higher-end vehicles
8	Fuel_Type_Hybrid	0.0252	Hybrid vehicles skew toward advanced tech adoption
9	Vehicle_Age	0.0249	Newer vehicles more likely to need modern security
10	Year	0.0248	Recency of manufacture affects available features

5. Insights and Recommendations Report

5.1 Key Insights

- Connected_Car and OTA_Update_Enabled are among the top 4 predictors - connectivity drives premium adoption
- OTA-enabled vehicles show 21.9% Premium Cyber Suite adoption vs only 0.3% for non-OTA vehicles
- ADAS_Level is the single strongest predictor - vehicles at ADAS Level 3 show 32.7% Premium adoption
- Luxury segment vehicles show the highest advanced-package adoption at 75.5%, followed by Premium (61.5%)
- Non-connected vehicles are almost entirely limited to Basic Monitoring or Threat Detection (87.9% combined)
- Threat Detection is the most commonly adopted package overall (23.9% of all vehicles)
- Hyundai leads Premium Cyber Suite adoption among brands at 11.8%, closely followed by Kia and Mahindra

5.2 Business Recommendations

- **1. Bundle Premium Cybersecurity with Connectivity Features** - Connected_Car and OTA_Update_Enabled are top drivers; bundle premium security with these features at point of sale
- **2. Prioritize High-ADAS Vehicles** - ADAS_Level is the strongest predictor; target Level 2-3 vehicles with Premium Cyber Suite offers
- **3. Segment-Based Marketing** - Luxury and Premium segments show highest advanced adoption; prioritize these segments for upsell campaigns
- **4. Basic Package for Entry Segment** - Non-connected, low-ADAS Budget vehicles are well served by Basic Monitoring / Threat Detection tiers
- **5. Brand-Level Targeting** - Hyundai, Kia, and Mahindra show the highest Premium Cyber Suite adoption; prioritize partnership discussions with these brands

5.3 Prediction Examples

Vehicle	Key Attributes	Predicted Package	Confidence
Vehicle A	Hyundai, Hybrid, ₹22,00,000, ADAS=2, Connected=Yes, OTA=Yes	Premium Cyber Suite	100%
Vehicle B	Tata, Petrol, ₹8,50,000, ADAS=0, Connected=No, OTA=No	Vulnerability Management	100%

5.4 Model Usage

Step	Description
Input	Vehicle feature data (14 input columns after preprocessing)
Process	Run through the trained best model (Decision Tree, selected by F1-Score)
Output	Predicted cybersecurity package with probability for each of the 5 classes
Use Case	Product tier design, dealership upsell recommendations, marketing segmentation

6. Conclusion

This project developed an end-to-end Automotive Cybersecurity Service Prediction system using a synthetic automotive dataset (3,000 vehicles, 15 columns). The workflow included data preparation, exploratory analysis, feature engineering, model training, hyperparameter tuning, and evaluation. Three multi-class classification models were compared using weighted F1-Score to address class imbalance across five cybersecurity packages, with Decision Tree automatically selected as the best model (weighted F1-Score = 0.978). Key insights identified connectivity features (Connected_Car, OTA_Update_Enabled), ADAS sophistication, and Price as the major drivers of cybersecurity package adoption, giving automotive OEMs and product teams a clear, data-driven basis for cybersecurity product strategy and targeted marketing.